Crossings never help: the Kobon conventions coincide, with certified values and the status of $K(14)$

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Abstract

The Kobon triangle problem asks for the maximum number of non-overlapping triangles determined by n straight lines. Three quantities appear in the literature under one name: $a_3^s(n)$, triangular faces of *simple* arrangements; $K(n)$, triangular faces of *arbitrary* arrangements (the object of Clément–Bader's definition, in which a counted triangle is "realized by 3 straight line segments"); and $K_{\mathrm{gen}}(n)$, the maximum size of a family of triangles with pairwise disjoint interiors in which other lines *may* cross a counted triangle, so that a counted side may be subdivided. Trivially $a_3^s \le K \le K_{\mathrm{gen}}$, and the classical segment-counting arguments bound only the first two.

Our main result is that the top two coincide: *crossings never help.* We prove a corner-descent lemma — every triangle bounded by three lines of an arrangement contains a triangular *face* of that arrangement — and deduce by a one-step exchange that $\mathrm{face}(\mathcal{A}) = \mathrm{broad}(\mathcal{A})$ for *every* arrangement, hence $K_{\mathrm{gen}}(n) = K(n)$ for all n. The convention ambiguity is therefore harmless for values, and the non-simple segment bound of Felsner and Kriegel applies to interior-disjoint families with crossings. The collapse is pointwise false for *optima*: we exhibit arrangements, including an exact-rational 6-line arrangement attaining the record 7, whose maximum interior-disjoint family contains a crossed triangle.

The collapse also makes an all-degeneracy search sound: since crossings may be forbidden without loss of generality, a propositional encoding in which every parallel and multiple-point pattern is free decides the classical $K(n)$. We prove a capacity inequality valid with arbitrary parallels and multiple points,

$$3|S| + C + 2Q + \sum_p k_p(k_p - 4) \;\le\; n(n-2),$$

where C counts line/triangle interior incidences, together with an exact bounded-face budget; these drive the encoding. With them we certify $K(n)$ for every $n \le 10$ — $2, 5, 7, 11, 15, 21, 25$ from $n = 4$, with `drat-trim`-verified refutations of $K(n)+1$ at $n = 4, \ldots, 9$ — admitting every degeneracy. These are the first machine-checkable all-degeneracy certificates known to us, but not the first claims of the values: web essays by Rasukaru in 2005 [10] give geometric arguments for $n = 6, 8, 10, 11$. At $n = 8$ the value 15 exceeds the exact simple maximum 14, so it is attained only non-simply; the certificate verifies precisely the regime where a simple-arrangement bound cannot apply. The same mismatch affects $n = 12$ and $n = 14$, where the constructions 38 and 54 exceed the exact simple maxima 37 and 53; we show the exactness claims recorded for them are unsupported. On the constructive side we verify, in exact rational arithmetic and with three implementations, a 14-line arrangement with 54 pairwise interior-disjoint triangles, and prove that every currently known extremal arrangement at $n = 11, 14, 18, 20$ attains its published count as its exact optimum. Finally we observe that the entry $a(14) = 54$ recorded as exact in OEIS A006066 on 21 August 2026 pairs a non-simple construction with a simple-arrangement upper bound; the rigorous status is $54 \le K(14) \le 55$, and that the same reading leaves $38 \le K(12) \le 39$, so

$n = 12$ is in fact the smallest open case. Each is closed by a single crossing-free instance with all degeneracies free. We also prove two equality-branch obstruction theorems ($n = 14$ and $n = 18$, machine-checked) and that the method cannot close the analogous branch at $n = 20$. Finally, we refute a natural face-specific square penalty suggested by sequential line addition: the classical simplicial family formed by the sides and symmetry axes of a regular polygon gives infinitely many straight-line counterexamples.

1 Three quantities

Let $\mathcal{A}$ be an arrangement of $n \geq 3$ distinct lines in the real affine plane, not all parallel (*essential*). A *segment* of $\mathcal{A}$ is a bounded maximal piece of a line between two consecutive crossing points.

Definition 1. (i) $a_3^s(n)$: the maximum number of triangular bounded faces of a *simple* arrangement of n lines (no two parallel, no three concurrent).

(ii) $K(n)$: the maximum, over all arrangements of n lines, of the number of triangles each of whose three sides is a single segment of $\mathcal{A}$, no two of them overlapping. Equivalently — since a line entering the interior of a triangle must cross two of its sides and thereby subdivide them — the maximum number of triangular bounded faces of an arbitrary arrangement.

(iii) $K_{\text{gen}}(n)$: the maximum, over all arrangements, of the size of a family S of triangles bounded by triples of lines of $\mathcal{A}$ whose open interiors are pairwise disjoint. Members of S may be crossed by other lines of $\mathcal{A}$.

Clearly $a_3^s(n) \leq K(n) \leq K_{\text{gen}}(n)$.

Clément and Bader [3] define a Kobon triangle as "a triangle that is realized by 3 straight lines segments and which does not overlap with other triangles", i.e. they work with K. Their mod-6 claim

$$K(n) \;\leq\; \lfloor n(n-2)/3 \rfloor - \mathbf{1}_{n \bmod 6 \in \{0,2\}} \tag{1}$$

passes through *perfect configurations*, defined as pairwise-intersecting arrangements in which every segment is the side of exactly one triangle. The load-bearing step in their Lemma 1 says that every multipoint belongs to at most two pairs of triangular faces sharing a side. This is false even in their face convention: Proposition 12 constructs a k-fold point belonging to all $2k$ possible shared radial sides, already six at a triple point and eight at a quadruple point. Thus the draft does not establish that equality forces a perfect configuration. This does not refute (1); it leaves the claimed bound unproved by the argument as written. Rasukaru's earlier web analysis [10], which is credited on the Honma page cited as reference [6] of [3], uses a different sequential charge: each additional line through an existing intersection loses at least three new segments and is assigned at most two new shared-side credits. This suggests the face-specific square penalty discussed below. Proposition 15 gives infinitely many straight-line counterexamples to that penalty, so the sequential credits cannot be globalised in this form. This does not by itself refute the resulting mod-6 numerical bound. The final generalisation on the web explicitly leaves two perturbation reductions at "probably harmless", so we treat it as prior insight rather than a proved theorem. The draft is additionally unpublished. Bartholdi, Blanc and Loisel [1] and Blanc [2] bound a_3^s. Thus:

> *The base segment bound $\lfloor n(n-2)/3 \rfloor$ is stated for K including non-simple arrangements as Theorem 2(3) of Felsner and Kriegel [8] (so used, e.g., in [12]), with the proof deferred there to Roudneff [9]; the mod-6 improvement (1) for K rests on the unpublished draft [3]; the sharpest numerical bounds [1, 2] are proved for a_3^s only and hence do not apply to*

non-simple record arrangements. No bound had been proved for K_{gen}; Theorem 4 below shows none is needed, since $K_{\text{gen}} = K$.

Remark 2 (provenance of the base bound)**.** For *simple* Euclidean arrangements the base bound is a two-line count: there are $n(n-2)$ bounded edges, and in a simple arrangement only one of the two cells along an edge can be a triangle (Grünbaum; equivalently Lemmas 5–6 of [8]), so $3p_3 \le n(n-2)$. For arbitrary arrangements the situation is more delicate — an edge with a multiple endpoint *can* border two triangular cells — and [8] states the bound as Theorem 2(3) for every Euclidean pseudoline arrangement while referring the proof to [9], whose theorem is the projective bound $p_3 \le n(n-1)/3$ for arbitrary arrangements with $n \ge 9$ (false for $n \le 8$). The reduction is not immediate in the direction needed: adjoining the line at infinity to a Euclidean arrangement of n lines and applying the projective bound gives only $p_3 \le n(n+1)/3 - 3$, because Levi's theorem guarantees just three triangles along the added line whereas $n(n-2)/3$ requires n of them. We therefore cite Theorem 2(3) as the refereed statement and record that the numerical upper ends of Table 2 inherit that provenance — one more reason to prefer the self-contained certificates of §7, whose unsatisfiability verdicts bound K with every degeneracy free and cite nothing.

2 Crossings never help

Write face($\mathcal{A}$) for the number of triangular bounded faces of $\mathcal{A}$ (distinct faces have disjoint interiors, so this is also the largest interior-disjoint family of uncrossed triangles) and broad($\mathcal{A}$) for the largest family of triangles bounded by triples of lines of $\mathcal{A}$ with pairwise disjoint interiors, crossings permitted. Then $K(n) = \max_{\mathcal{A}} \operatorname{face}(\mathcal{A})$ and $K_{\text{gen}}(n) = \max_{\mathcal{A}} \operatorname{broad}(\mathcal{A})$, and face $\le$ broad pointwise.

Lemma 3 (corner descent)**.** *Let T be a nondegenerate triangle bounded by three lines of $\mathcal{A}$. Then T contains a triangular face of $\mathcal{A}$.*

Proof. Induct on the number k of lines of $\mathcal{A}$ meeting $\operatorname{int} T$. If $k = 0$ then T is itself a face. Let $k \ge 1$ and let c meet $\operatorname{int} T$. The chord $c \cap T$ has its two endpoints on ∂T, and c contains at most one vertex of T: two vertices of T would lie on a common side line, forcing c to be that line. Two cases.

If c contains no vertex, its endpoints lie in the relative interiors of two distinct sides, on lines ℓ_a, ℓ_b say, and c cuts off the corner $\ell_a \cap \ell_b$: the corner piece is the triangle on $\{\ell_a, \ell_b, c\}$, nondegenerate because c misses that vertex.

If c contains the vertex $V = \ell_a \cap \ell_b$, its other endpoint lies in the relative interior of the opposite side, on ℓ_{opp}, and the two pieces are the triangles on $\{\ell_a, \ell_{\text{opp}}, c\}$ and $\{\ell_b, \ell_{\text{opp}}, c\}$; each is nondegenerate, since $c \cap \ell_a = V \notin \ell_{\text{opp}}$.

In both cases some piece $T' \subseteq T$ is a nondegenerate triangle bounded by three lines of $\mathcal{A}$, and every line meeting $\operatorname{int} T'$ meets $\operatorname{int} T$ while c does not meet $\operatorname{int} T'$; so the induction parameter of T' is at most $k - 1$. Applying the inductive hypothesis to T' produces a triangular face inside $T' \subseteq T$. □

Theorem 4 (collapse)**.** face($\mathcal{A}$) = broad($\mathcal{A}$) *for every arrangement $\mathcal{A}$. Consequently $K_{\text{gen}}(n) = K(n)$ for every n.*

Proof. Let S be an interior-disjoint family of triangles of $\mathcal{A}$ with $|S| = \operatorname{broad}(\mathcal{A})$, and suppose $T \in S$ is crossed. By Lemma 3 there is a triangular face $F \subseteq T$; then $\operatorname{int} F \subseteq \operatorname{int} T$, so $S \setminus \{T\} \cup \{F\}$ is again interior-disjoint, has the same cardinality, and has one fewer crossed member. Iterating at most $|S|$ times yields an interior-disjoint family of uncrossed triangles, i.e. of triangular faces, of size broad($\mathcal{A}$). Hence face($\mathcal{A}$) $\ge$ broad($\mathcal{A}$), and the reverse inequality is trivial. Taking maxima over $\mathcal{A}$ gives $K_{\text{gen}}(n) = K(n)$. □

Corollary 5. *$K_{\mathrm{gen}}(n) \le \lfloor n(n-2)/3 \rfloor$, and $K_{\mathrm{gen}}(n) \le \lfloor n(n-2)/3 \rfloor - \mathbf{1}_{n \bmod 6 \in \{0,2\}}$ if (1) is granted. In particular the three quantities of Definition 1 satisfy $a_3^s \le K = K_{\mathrm{gen}}$.*

We could not locate either statement in the literature. The closest published relative is Lemma 4 of Leaños, Mbe Koua and Rivera [5], which reaches the same conclusion for *simple* arrangements under the extra hypothesis that one side of T is uncrossed, by a chord induction like the one above; the all-sides-crossed case is not covered there. The convention itself is flagged in only two places we found: Zamfirescu [7] defines *facial* triangles ($L \cap \operatorname{int} \Delta = \emptyset$ for all $L \in \mathcal{A}$) alongside general ones, and the Kobon variations recorded after Forge and Ramírez Alfonsín [4] ask that counted triangles not be crossed. Neither remarks that the count is unaffected. That the collapse is not automatic is shown by Zamfirescu's own problem: for *congruent* triangles the two conventions genuinely differ, her facial count being 14 at $n = 7$ against a larger general count [7]. The exchange of Theorem 4 uses interior-disjointness, which a congruence constraint destroys.

Two remarks delimit the result. First, the collapse is a statement about maxima, not about individual optima: an optimal family may genuinely contain a crossed triangle. For $n = 4$ this is typical rather than exceptional — in 296 of 300 random exact-rational generic arrangements some optimum of size $K(4) = 2$ uses a crossed triangle — and at $n = 6$ the arrangement

$$y = -\tfrac{9}{2}x - \tfrac{9}{2}, \quad y = -4x - 4, \quad y = -\tfrac{18}{5}x - 3, \quad y = -\tfrac{19}{6}x - \tfrac{19}{6}, \quad y = \tfrac{13}{3}x + 7, \quad y = 5x + \tfrac{4}{3}$$

(lines $0, 1, 3$ concurrent, no parallels) has face $=$ broad $= 7 = K(6)$ with the interior-disjoint optimum $\{012, 015, 024, 134, 135, 235, 245\}$ in which 012 is crossed by line 4; descent replaces 012 by the face 014 and recovers the face optimum, which is exactly the exchange of Theorem 4. Second, the descent is effective, and both statements have been machine-checked on exact-rational arrangements with planted parallel classes, triple points, quadruple points and pairs of triple points: over 872 arrangements at $n \le 7$ all 13,017 triangles descend to a face (depths up to 4) and no arrangement has face $<$ broad (`scratch/kobon/collapse_verify.py`).

Remark 6. Theorem 4 makes a search restriction sound: "some arrangement of n lines carries T pairwise interior-disjoint triangles" is equivalent to "some arrangement of n lines carries T triangles none of which is crossed". Forbidding crossings is therefore a WLOG, and it may be imposed on top of an encoding in which every parallel and multiple-point pattern remains free; §7 uses this.

This has an immediate consequence for the current record. On 21 August 2026 OEIS A006066 [16] was updated to list $a(14) = 54$ as exact, attributed to Maiorana's arrangement [14]. That arrangement is *not* simple: it has two triple points (on lines $\{0, 4, 10\}$ and $\{0, 11, 12\}$, sharing line 0) and no parallels. The matching upper bound 54 in the same table is the BBL value $\lfloor n(n - 7/3)/3 \rfloor$, proved for simple arrangements. For K, the peer-reviewed non-simple segment bound [8] gives 56 at $n = 14$, and the mod-6 improvement (1) of the draft [3] gives 55. Hence:

Proposition 7. *With the published literature, $54 \le K(14) \le 56$ unconditionally, and $54 \le K(14) \le 55$ if the draft [3] is granted; by Theorem 4 the same window holds for $K_{\mathrm{gen}}(14)$. In every case $a(14) = 54$ is a lower bound paired with a simple-arrangement upper bound, so its exactness is not established.*

Contributions.

1. The collapse $K_{\mathrm{gen}} = K$ (Theorem 4), from a corner-descent lemma, with the search WLOG it licenses (Remark 6) and exact-rational witnesses showing the corresponding pointwise statement is false.

2. A capacity theorem for K_{gen} (Theorem 16), with exact degeneracy penalty and crossing charge, and a closed-form ceiling (Theorem 20).

3. Exact edge-use and sector-run identities, an infinite family of straight-line counterexamples to the tempting coefficient-1 square penalty, and a checked finite certificate for the surviving coefficient-2/3 candidate at $n = 7$ (Lemma 13, Corollary 14, and Propositions 15–26).

4. A certified initial segment with all degeneracies free: $K(4), \ldots, K(9) = 2, 5, 7, 11, 15, 21$, each with a `drat-trim`-verified refutation of $K(n) + 1$, plus $K(10) = 25$ from a prior cube cover and an independent verified face-monolith refutation (Theorem 27); the $n = 6, 8, 10$ rows are machine-checkable all-degeneracy certifications of values argued on the web in 2005 (Remark 9).

5. A hypothesis audit of the recorded small values (Proposition 8): at $n = 8, 12, 14$ the record exceeds the exact simple maximum, so the simple-only bound cited as "the upper bound" cannot apply.

6. An independently verified $K_{\text{gen}}(14) \geq 54$ (Theorem 28) and an exact optimality census of all known extremal arrangements (Theorem 30), which shows $K = K_{\text{gen}}$ on each of them.

7. Two equality-branch obstruction theorems with machine-checked finite certificates (Theorems 31, 32) and a proof that the same method cannot close the analogous branch at $n = 20$ (Proposition 33).

8. The status audit of $K(14)$ (Proposition 7) and the decision procedure that closes it.

3 Notation

For $\ell \in \mathcal{A}$ let q_ℓ be the number of lines parallel to ℓ and $Q = \frac{1}{2}\sum_\ell q_\ell$. A *multipoint* has multiplicity $k_p \geq 3$; N_k counts points of multiplicity k. On ℓ, r_ℓ is the number of distinct crossings, a_ℓ the number of multipoints, and $m_\ell = r_\ell - 1$ the number of bounded elementary gaps. For an admissible family S (Definition 1(iii)) set

$$\sigma_\ell = \#\{T \in S : \ \ell \text{ supports a side of } T\}, \qquad \gamma_\ell = \#\{T \in S : \ \ell \cap \operatorname{int}(T) \neq \emptyset\},$$

so $\sum_\ell \sigma_\ell = 3|S|$ and $C := \sum_\ell \gamma_\ell$ is the total number of line/triangle interior incidences. Write $T = |S|$ and $\Delta = n(n-2) - 3T$.

4 What is actually proved for small n

The recorded values are widely treated as exact. They are not, and the reason is a hypothesis mismatch that Theorem 4 lets us state cleanly, since only one convention now exists.

Write $\bar{a}_3^s(n)$ for the exact maximum over *simple* affine arrangements. Blanc [2] (Theorem 1 for the bound, Theorem 3 for attainment, computer-assisted) and [1] (Theorem 1.4) give $\bar{a}_3^s(8) = 14$, $\bar{a}_3^s(11) = 32$, $\bar{a}_3^s(12) = 37$ and $\bar{a}_3^s(14) = 53$. Meanwhile the even-n bound $\lfloor n(n - 7/3)/3 \rfloor$ of [1, Thm. 1.1] — proved under the explicit hypothesis "our sole interest is with simple arrangements, i.e. arrangements without multiple intersections" — takes the values $7, 15, 25, 38, 54$ at $n = 6, 8, 10, 12, 14$, which are exactly the recorded constructions.

Proposition 8. *For $n \in \{8, 12, 14\}$ every arrangement attaining the recorded value (15, 38, 54 respectively) is non-simple, because that value exceeds $\bar{a}_3^s(n) \in \{14, 37, 53\}$. Consequently the simple-arrangement bound $\lfloor n(n-7/3)/3 \rfloor$ of [1, Thm. 1.1], which coincides with the recorded value at those n, cannot be applied to the arrangements in question, and the exactness claims that invoke it — the upper-bound column of OEIS A006066 [16] at $n = 12$ and its comment of 13 August 2026 at $n = 14$, and the corresponding table in MathWorld [15] — are unsupported. The published bounds valid for arbitrary arrangements are $\lfloor n(n-2)/3 \rfloor = 16, 40, 56$ [8], and $15, 39, 55$ if the unpublished draft [3] is granted.*

Proof. A simple arrangement has at most $\bar{a}_3^s(n)$ triangular faces by definition of $\bar{a}_3^s$, and $15 > 14$, $38 > 37$, $54 > 53$. The remaining statements are the cited hypotheses and values. □

At $n = 6$ and $n = 10$ the construction equals $\bar{a}_3^s(n)$, so it may be simple; but the same gap remains on the upper side, since $\lfloor n(n-2)/3 \rfloor$ is 8 and 26 there. Collecting:

Remark 9 (source status)**.** The refereed literature before this work determines $K(n)$ for $n \le 5$ and $n \in \{7, 9, 13\}$, where $\lfloor n(n-2)/3 \rfloor$ is attained, together with $K(11) = 32$ [11]. This is not the whole prior record. Rasukaru's public Japanese web essays of September–October 2005 [10], preserved and credited by Honma, give geometric arguments for $K(6) = 7$, $K(8) = 15$, $K(10) = 25$ and $K(11) = 32$, and state a general even-n bound that gives 54 at $n = 14$. The final generalisation explicitly treats two perturbation reductions as "probably" harmless, and the work is neither refereed nor accompanied by a checkable certificate. Accordingly, Theorem 27 claims machine-checkable certificates, not priority for the numerical values.

5 The capacity theorem

Lemma 10 (incidence identity)**.** *$m_\ell + 2a_\ell = n - 2 - q_\ell - \sum_{p \ni \ell}(k_p - 4)$ for every $\ell \in \mathcal{A}$.*

Proof. The $n - 1 - q_\ell$ lines meeting ℓ do so in r_ℓ distinct points, and a multipoint p on ℓ absorbs $k_p - 1$ of them into one point, so $n - 1 - q_\ell = r_\ell + \sum_{p \ni \ell}(k_p - 2)$. Hence $m_\ell + 2a_\ell = r_\ell - 1 + 2a_\ell = n - 2 - q_\ell - \sum_{p \ni \ell}(k_p - 2) + 2a_\ell$, and a_ℓ is the number of summands. □

Lemma 11 (per-line capacity)**.** *$\sigma_\ell + \gamma_\ell \le m_\ell + 2a_\ell$ for every $\ell \in \mathcal{A}$ and every admissible S.*

Proof sketch. Each $T \in S$ with a side on ℓ determines the side interval $I_T \subset \ell$; each T crossed by ℓ determines the chord interval $J_T = \ell \cap \operatorname{int}(T)$. Disjointness of interiors makes the J_T pairwise disjoint, makes each J_T disjoint from the interiors of side intervals on the same side of ℓ, and makes side intervals on a fixed side of ℓ interior-disjoint. Every such interval contains at least one elementary gap, and two intervals can claim the same gap only when they lie on opposite sides of ℓ and abut a common multipoint of ℓ, each of which admits at most two coincidences. Charging one gap per interval and two tokens per multipoint yields the claim; the exhaustive case analysis is carried out in the companion document [17, §3] and was re-derived by an adversarial audit over 3,016 arrangements and 812,486 admissible families with 5,900,863 exact linewise checks. □

Proposition 12 (flower saturation)**.** *For every $k \ge 3$ there is an arrangement of $k + 3$ straight lines with a k-fold point p incident to $2k$ bounded triangular faces. Every one of the $2k$ elementary radial segments from p is the common side of the two triangular faces beside it. Consequently the two tokens per incident line in Lemma 11 can be simultaneously realised at one multipoint, and the local coefficient $k(k-4)$ in (2) is sharp.*

Proof. Choose a triangle Δ, a point $p \in \operatorname{int}(\Delta)$, and k distinct lines through p, three of which pass through the three vertices of Δ. Choose the remaining lines generically. The $2k$ rays cut the boundary of Δ cyclically; because every vertex of Δ lies on a ray, the boundary arc between consecutive rays is contained in one side of Δ. Each of the $2k$ sectors inside Δ is therefore a triangular arrangement face. Each radial segment is adjacent to the sectors on both sides and hence is used twice.

Collapsing k pairwise crossings to p removes $k-2$ elementary gaps on each of the k incident lines, or $k(k-2)$ in total, while the $2k$ shared radial segments return $2k$ units. The net local charge is $k(k-2) - 2k = k(k-4)$. Exact rational instances for $k = 3, 4$, including all face and shared-side checks, are replayed by `scratch/kobon/flower_counterexamples.py`. $\square$

Lemma 13 (edge-use identity)**.** *Let F be the number of bounded triangular faces, M the number of bounded elementary segments, E_2 the number incident with a triangular face on both sides, and U the number incident with no triangular face. Then*

$$3F = M + E_2 - U, \qquad M = n(n-2) - 2Q - \sum_p k_p(k_p - 2).$$

Consequently the proposed square penalty

$$3F \le n(n-2) - 2Q - \sum_p (k_p - 2)^2$$

is equivalent to

$$E_2 - U \le 2 \sum_p (k_p - 2).$$

Proof. If E_1 counts bounded elementary segments incident with exactly one triangular face, then $3F = E_1 + 2E_2$ and $M = U + E_1 + E_2$. On a line ℓ there are

$$m_\ell = n - 2 - q_\ell - \sum_{p \ni \ell} (k_p - 2)$$

bounded elementary segments. Sum over ℓ and use $\sum_\ell q_\ell = 2Q$. Finally, $k_p(k_p-2) - (k_p-2)^2 = 2(k_p - 2)$. $\square$

Corollary 14 (sector-run excess)**.** *At a k_p-fold point p, its $2k_p$ rays determine $2k_p$ angular sectors, each lying in one incident cell; call a sector triangular when that cell is a triangular face with vertex p. Let z_p be the number of non-triangular sectors and r_p the number of cyclic runs of triangular sectors, with $r_p = 0$ when $z_p = 0$. Let D_2 count doubly used segments whose two endpoints are both multipoints. Then*

$$3F - \left(n(n-2) - 2Q - \sum_p (k_p - 2)^2 \right) = \sum_p (4 - z_p - r_p) - U - D_2.$$

Proof. A segment $e \subset \ell$ used by triangular faces on both sides has at least one multipoint endpoint. Otherwise let a and b be the unique crossing lines at its two simple endpoints. At either endpoint the boundary of each adjacent face must turn from ℓ onto the crossing line; continuing along ℓ would not bound that face. Thus both triangles have the same three supporting lines ℓ, a, b. If a and b are parallel neither triangle closes; otherwise they have one intersection and the three lines determine only one bounded triangle, not triangles on opposite sides of e.

At p, a ray begins a doubly used segment exactly when both adjacent sectors are triangular. If $z_p = 0$ all $2k_p$ rays qualify. Otherwise a run of length s contributes $s-1$, so the number of qualifying rays is $I_p = 2k_p - z_p - r_p$. Summing I_p counts each doubly used segment once, and once more exactly when both endpoints are multipoints; hence $\sum_p I_p = E_2 + D_2$. Therefore

$$E_2 - U - 2\sum_p (k_p - 2) = \sum_p (4 - z_p - r_p) - U - D_2,$$

and Lemma 13 identifies the left side with the claimed excess. □

Proposition 15 (the square penalty fails)**.** *The face-specific square penalty of Lemma 13 is false for straight-line arrangements. In fact it fails for infinitely many values of n. There is an exact rational* 12*-line counterexample with no parallels, $N_3 = 15$, $N_6 = 1$, and all* 30 *bounded faces triangular:*

$$3F = 90 > 120 - \big(15 + (6-2)^2\big) = 89.$$

Proof. Let $\mathcal{R}_m$ be Grünbaum's regular simplicial projective arrangement $A(2m, 1)$ in the family $R(1)$: the m side lines and the m mirror axes of a regular m-gon [6, pp. 2, 4]. For even m, half the axes join opposite vertices and half join opposite edge midpoints; for odd m, each axis joins a vertex to the opposite edge midpoint. In either case the m axes are distinct. The catalogue gives, for $m > 3$, one m-fold centre, $\binom{m}{2}$ triple points and m double points; our counterexample range is $m \geq 6$. Directly, every pair of side lines meets on the symmetry axis interchanging them, and these points are distinct because no three tangents to a conic are concurrent. These triples consume $m(m-1)$ of the m^2 side–axis pairs, leaving m double points, while all axis–axis pairs meet at the centre.

Thus

$$f_0 = 1 + \binom{m}{2} + m = \frac{m^2 + m + 2}{2}.$$

Projective Euler and simpliciality give $f_2 = 2(f_0 - 1) = m^2 + m$. Choose a generic projective line as the line at infinity. It avoids every arrangement vertex, so the resulting affine arrangement has $Q = 0$ and the same finite multiplicities. Its $2m$ intersections with the arrangement divide the line at infinity into $2m$ arcs, so at most $2m$ projective chambers become unbounded. Hence at least

$$F \geq f_2 - 2m = m(m-1)$$

triangular faces remain bounded. The square correction is

$$\binom{m}{2} + (m-2)^2.$$

With $n = 2m$, the excess of $3F$ over the proposed right-hand side is therefore at least

$$3m(m-1) - \left(2m(2m-2) - \binom{m}{2} - (m-2)^2\right) = \frac{m^2 - 7m + 8}{2},$$

which is positive for every $m \geq 6$. For $m = 6$, the exact chart replay cited in Appendix A finds $F = 30$, the stated census, 39 doubly used and no unused bounded segments, and verifies both sides as 90 and 89 by two independent face predicates. Its sector-run census gives $\sum_p (4 - z_p - r_p) = 27$ and $D_2 = 26$, exposing the one-unit violation in Corollary 14. □

Theorem 16 (capacity)**.** *For every essential arrangement of n distinct affine lines and every admissible family S,*

$$3|S| + C + 2Q + \sum_p k_p(k_p - 4) \;\le\; n(n-2). \tag{2}$$

Proof. Sum Lemma 11 over all lines and substitute Lemma 10; the left side gives $3|S| + C$, and on the right $\sum_\ell q_\ell = 2Q$ while each multipoint lies on exactly k_p lines, so $\sum_\ell \sum_{p \ni \ell}(k_p - 4) = \sum_p k_p(k_p - 4)$. □

Setting $C = 0$ recovers, with explicit degeneracy corrections, the classical segment bound for K; the point of (2) is that the crossing charge makes it valid for K_{gen}.

Remark 17. Neither correction term can be replaced by the naive alternatives: the variants with $\binom{k_p-2}{2}$ or $(k_p-2)^2$ in place of $k_p(k_p-4)$ are false, with exact counterexamples in [17, §6]. The all-parallel class is the unique inessential exception.

Corollary 18. *If $\mathcal{A}$ has no triple point then $|S| \le \frac{1}{3}\big(n(n-2) - C - 2Q - \sum_{k\ge 4} N_k k(k-4)\big) \le n(n-2)/3$. In general $|S| \le n(n-2)/3 + N_3 - \frac{1}{3}\big(C + 2Q + \sum_{k \ge 5} N_k k(k-4)\big)$, so triple points are the only degeneracy that can raise the ceiling, by at most one triangle each.*

6 A closed-form ceiling for K_{gen}

Lemma 19 (face budget)**.** $|S| + C + Q + \sum_p \binom{k_p-1}{2} \le \binom{n-1}{2}$.

Proof. $\mathcal{A}$ has exactly $\binom{n-1}{2} - Q - \sum_p \binom{k_p-1}{2}$ bounded faces. The interior of $T \in S$ contains at least $1 + c_T$ bounded faces, where c_T is the number of lines crossing it, and distinct members have disjoint interiors, so they consume disjoint face sets; summing gives $|S| + C$. □

Theorem 20 (closed-form ceiling)**.** $K_{\mathrm{gen}}(n) \le \lfloor (n-2)(5n-3)/12 \rfloor$ *for all $n \ge 3$. In the combined inequality used, multiplicity* 3 *has coefficient exactly* 0 *and every $k \ge 4$ has coefficient* $k(k-4) + 3\binom{k-1}{2} > 0$.

Remark 21. Since $5n - 3 > 4n$ for $n > 3$, this ceiling is weaker than $\lfloor n(n-2)/3 \rfloor$ at every $n \ge 4$, so Corollary 5 supersedes it numerically. We retain it because it is proved directly from the capacity inequality and the face budget without the exchange argument, and because those two inequalities — not the closed form — are what the encoding of §7 propagates.

Proof. Add (2) to three times Lemma 19:

$$6|S| + 4C + 5Q + \sum_{k \ge 3} N_k\Big(k(k-4) + 3\binom{k-1}{2}\Big) \le n(n-2) + 3\binom{n-1}{2} = \frac{(n-2)(5n-3)}{2}.$$

The bracket is $0, 9, 23, 42, \ldots$ for $k = 3, 4, 5, 6$, hence nonnegative. □

Lemma 22 (reduction at zero slack)**.** *If $\Delta = 0$ and $\mathcal{A}$ has no triple point, then $C = 0$, $Q = 0$, every multipoint has $k_p = 4$, and every member of S is a bounded triangular face; in particular $|S| \le K(n)$ is attained by faces.*

Proof. (2) forces $C + 2Q + \sum_{k \ge 4} N_k k(k-4) \le 0$ with all terms nonnegative. □

Remark 23 (zero slack is not simplicity)**.** Lemma 22 deliberately does not conclude that the arrangement is simple: quadruple points have zero coefficient in (2). Proposition 12 shows that their local double-use credit is real, not an artefact of the proof. Any implication "$\Delta = 0$ forces simple" therefore needs a genuinely stronger global inequality; it does not follow from the capacity theorem.

Among known values, (2) is attained exactly at $n = 3, 5, 9, 15, 17$ — precisely the n where the Tamura value $n(n-2)/3$ is realised — and there all three quantities of Definition 1 coincide.

Lemma 24 (direct gap criterion)**.** *Let $t = \{a, b, c\}$ be a nondegenerate triple in an arbitrary straight-line arrangement, allowing parallels and multiple points elsewhere. Then t bounds a triangular arrangement face if and only if, for each side line $r \in t$, no outside line $u \notin t$ meets r strictly between the two vertices of t on r.*

Proof. Necessity is immediate: such an intersection subdivides the relative interior of a face side. Conversely, if an outside line enters the open triangle, it must leave through the boundary. It cannot meet two vertices, since the line through two vertices is one of the three side lines; if it enters through one vertex, it leaves through the relative interior of the opposite side. Otherwise it meets two side interiors. In every case some forbidden strict between-intersection exists. Likewise an arrangement vertex in the open triangle would put each of its incident lines through the boundary. Thus the three unsubdivided sides bound one arrangement cell. □

In the slope-sorted order encoding, the forbidden event on side r with endpoint lines a, b is exactly

$$S(t) \wedge X(r, a, u) \wedge X(r, u, b) \quad \text{or} \quad S(t) \wedge X(r, b, u) \wedge X(r, u, a).$$

Concurrency at an endpoint makes both strict patterns false, as required. Distinct arrangement faces have disjoint interiors, so no pairwise polygon disjointness clauses are needed once Lemma 24 is imposed.

Lemma 25 (hereditary, sector, endpoint, and perturbation cuts)**.** *Let S be a family of triangular arrangement faces.*

1. *For every subset $W \subseteq \mathcal{A}$, the members of S whose three supporting lines lie in W are distinct triangular faces of the subarrangement W. In particular their number is at most $K(|W|)$.*
2. *At most four members of S use any fixed pair of lines. If that pair meets at a finite multipoint, at most two do. In the latter case the number is zero unless the pair is consecutive on at least one of the two arcs of the cyclic slope order restricted to the lines through the point.*
3. *Let distinct faces $\{r, a, b\}$ and $\{r, c, d\}$ share line r. If $P_{ra} = P_{rc}$ and P_{rb}, P_{rd} lie on the same ray of r from that point, then $P_{rb} = P_{rd}$. If both endpoint equalities hold, the faces share their side segment on r and their apexes P_{ab}, P_{cd} lie in opposite open halfplanes of r. This includes $a = c$, where the first equality is automatic; hence two faces sharing a line pair and no remote concurrency occupy opposite sectors.*
4. *Let $M(S)$ be the number of members of S having a vertex at a finite multipoint. Then*

$$|S| - M(S) \leq \bar{a}_3^s(n),$$

where $\bar{a}_3^s(n)$ is the simple-arrangement maximum. In particular, a 12-line family of 39 faces has at least two multipoint-incident members.

Proof. Deleting lines outside W removes no boundary line of a face supported in W and creates no new crossing, so its three unsubdivided sides still bound the same triangular cell. Distinct support triples remain distinct.

At a simple intersection a fixed pair bounds four open sectors, and a cell incident with the intersection occupies exactly one. At a k-fold point with $k \geq 3$, the $2k$ rays show that the fixed pair bounds a cell only when its rays are consecutive; then it bounds exactly two opposite sectors. In the standard chart with no vertical line, the labels are the linear slope order. Thus lines with labels strictly between the pair lie on one projective arc and labels outside the interval lie on the other, proving the last assertion.

For the endpoint statement, a face side has no arrangement vertex in its relative interior. Two distinct points on the same ray are linearly ordered, and the nearer would lie in the relative interior of the side ending at the farther. Thus the remote points coincide. If both endpoints coincide, the two triangular cells border the same arrangement edge; distinct cells incident with one edge lie on its two opposite sides, proving the apex assertion.

Finally retain only faces counted by $|S| - M(S)$. Their three vertices are simple, and the direct-gap inequalities defining every side are strict. There is therefore one sufficiently small generic perturbation of all line coefficients that preserves all these finitely many inequalities while removing every finite concurrency and parallelism. Intersections created from formerly parallel distinct lines escape to infinity as the perturbation tends to zero, so they do not enter the compact preserved sides. The retained faces survive in a simple arrangement and are at most $\bar{a}_3^s(n)$. At $n = 12$, $\bar{a}_3^s(12) = 37$, giving $M(S) \geq 39 - 37 = 2$. □

If exactly T faces are selected, the first cut is equivalently the lower bound $T - K(|W|)$ on selected faces meeting $\mathcal{A} \setminus W$. This complementary form gives substantially smaller cardinality counters near the frontier. For $|W| = 4$, the certified $K(4) = 2$ bound is cheaper in direct form: the four possible faces give four ternary prime clauses forbidding each three.

Proposition 26 (certified endpoint control at $n = 7$)**.** *Every essential arrangement of seven straight lines satisfies*

$$9F + 6Q + 2\sum_p (k_p - 2)^2 \leq 105,$$

equivalently the coefficient-$2/3$ *inequality in the second open problem below.*

Proof. Using Lemma 24, encode every selected face by the strict crossing orders on its three sides. Leave the selected-face count, all parallel classes and all finite concurrency patterns free, and impose only essentiality. A canonical first triple and the later lines through its point encode

$$(k-2)^2 = 1 + 3u + 2\binom{u}{2}, \qquad u = k - 3.$$

A strict violation is the integer score threshold 106. The resulting necessary order-table relaxation has 74,507 variables and 152,419 clauses. Kissat reports unsatisfiable; `drat-trim` verifies the 34,133,280-byte proof, with 59,128 formula clauses and 211,032 lemmas in the core. Every geometric violation extends this instance by selecting all of its triangular faces, so verified unsatisfiability proves the claim. □

7 Certified small values, all degeneracies free

The decision problem "is there an n-line arrangement with an admissible family of size T?" is encoded in propositional logic over slope-sorted combinatorial data: variables for crossing versus

n	$K_{\text{gen}}(n)$	refuted T	variables / clauses	core clauses	core lemmas	verify (s)
4	2	3	207 / 994	90	131	0.05
5	5	6	972 / 5,257	424	558	0.04
6	7	8	3,655 / 20,681	4,541	7,018	0.48
7	11	12	10,682 / 63,405	12,255	31,595	4.47
8	15	16	27,451 / 165,367	31,218	411,278	117.6
9	21	22	61,569 / 377,405	66,346	2,150,510	1060.3
10	25	26	131,314 / 792,462	52,808	40,395,481	58,458.4

Table 1: Certified initial segment. Every row is a `s VERIFIED` `drat-trim` run on a kissat proof of the refutation of $T = K_{\text{gen}}(n) + 1$; proofs range from 12 kB to 14.05 GB and the resolution counts from 582 to $3.46 \cdot 10^9$. Ledgers: `scratch/kobon/ladder_certificates.json` and `scratch/kobon/ladder_n10_t26_faces_certificate.json`.

parallel, concurrency of triples, the order of crossings along each line, the selection, and triangle/line crossing incidences, with the per-line inequalities of Lemma 11 and the budget of Lemma 19 as exact cardinality constraints. All degeneracies are free, so a verdict covers crossed and degenerate configurations simultaneously.

Theorem 27. *In the convention of Definition 1(iii),*

$$K_{\text{gen}}(4), \ldots, K_{\text{gen}}(9) \;=\; 2,\ 5,\ 7,\ 11,\ 15,\ 21,$$

and each value is certified: the instance $(n, K(n) + 1)$ is unsatisfiable with a `drat-trim`-verified proof (Table 1), and the matching lower bounds are the classical constructions. Together with the prior eleven-cube cover for $n = 10$ this gives $K(n)$ for all $n \leq 10$; the $n = 10$ value is confirmed independently through a different encoding by a single checked monolithic refutation of $(10, 26)$ in the crossing-free formulation of Theorem 4 ($131\,314$ variables, $792\,462$ clauses, 14,050,806,440-byte DRAT, `s VERIFIED`). The rungs at $n = 6$, $n = 8$ and $n = 10$ are the first machine-checkable all-degeneracy certificates known to us for those values. They independently verify the 2005 web arguments [10] and rule out 8, 16 and 26 triangles respectively without perturbation assumptions or a simple-arrangement hypothesis. At $n = 8$ this matters structurally: $K(8) = 15$ is attained only non-simply (Proposition 8).

These statements are strictly stronger than the corresponding classical ones, since they also exclude crossed families: e.g. $K_{\text{gen}}(8) = 15$ rules out the value $16 = \lfloor 8 \cdot 6/3 \rfloor$ left open for K_{gen} by the segment bound, and $K_{\text{gen}}(6) = 7$ rules out 8.

8 A verified lower certificate at $n = 14$

Theorem 28. $K_{\text{gen}}(14) \geq 54$*, witnessed by an arrangement with exact rational coefficients.*

Proof. From the SHA-256-pinned source file of [14] we verified: the 14 lines are pairwise non-proportional; the computed multipoint census is exactly two triple points sharing a line, with $Q = 0$, matching the file's declared data; exactly 54 triples have interiors met by no other line, computed with two independent predicates, the second of which also rejects the case of a line through a vertex separating the opposite side; and the resulting family passes the audited exact-rational checker `verify_selection`, which validates non-degeneracy and pairwise interior-disjointness in homogeneous rational coordinates. The same verification succeeds for all fifteen published solutions. □

9 Crossing incidences are exactly reified

The experiments that force or forbid crossings rest on a single literal being exact, so we record what it encodes. For a triple t of lines and a line $r \notin t$, let $\mathrm{TC}(t, r)$ be the variable defined by

$$\mathrm{TC}(t, r) \iff S(t) \wedge \Big(\bigvee_{e \subset t} \mathrm{SA}(e, r)\Big) \wedge \Big(\bigvee_{e \subset t} \mathrm{SB}(e, r)\Big), \tag{3}$$

where $S(t)$ says t is in the selected family and $\mathrm{SA}(e, r)$, $\mathrm{SB}(e, r)$ say that the vertex of t opposite[1] lies strictly above, respectively strictly below, r. Both sidedness literals are exact in the model.

Lemma 29. *In every arrangement,* $\mathrm{TC}(t, r)$ *holds precisely when t is selected and r meets the open interior of t.*

Proof. If some vertex of t is strictly above r and another strictly below, then r crosses the two sides joining them at relative interior points, and the open chord between those points lies in the open interior. Conversely, suppose r meets the open interior. Then r cannot contain two vertices of t: two vertices span a side, which lies on a line of t, and $r \notin t$. So at most one vertex is on r, and the chord separates the remaining two strictly, giving one strictly above and one strictly below. The conjunct $S(t)$ is asserted by the first clause of (3). □

The point of the lemma is that TC is attached to a *selected* triangle: a disjunction over TC literals asserts that the family exhibited by the model has a crossed member, not merely that the arrangement crosses some candidate triple. Cevians and lines through a vertex are handled by the "at most one vertex on r" case rather than excluded.

10 Exact optimality of the known witnesses

For a *fixed* arrangement, the largest admissible family is the independence number of the overlap graph on non-degenerate triples, with edges joining pairs of intersecting interiors. We compute it exactly: interiors are compared in homogeneous rational arithmetic, and optimality is certified by SAT (a family of size K is a model, UNSAT at $K + 1$ certifies the optimum).

Theorem 30. *For each of the following the exact admissible optimum equals the published count, with certified optimality: the fifteen 14-line 54-triangle arrangements of [14]; two exact rational 53-triangle 14-line reconstructions; an 11-line 32-triangle arrangement; an 18-line 93-triangle reconstruction; and a 20-line 116-triangle reconstruction. In particular $K = K_{\mathrm{gen}}$ on every known extremal arrangement.*

By Lemma 29 we may also ask whether an *optimal* family must be crossing-free, by conjoining $\bigvee_{t,r} \mathrm{TC}(t, r)$ with exact selection of $K(n)$ triangles. This is unsatisfiable for $n \in \{5, 7, 8, 9\}$ and satisfiable for $n \in \{4, 6\}$, so crossed optima exist but only at the smallest sizes; at $n = 4$ they are generic, occurring in 296 of 300 random exact-rational arrangements, and at $n = 6$ an explicit arrangement carries both a crossed and an uncrossed optimal family of size 7. The satisfiable cases are therefore not artefacts of degeneracy, and by Theorem 4 they never increase the optimum.

Consequently any improvement on the records must come from new arrangements, and by Corollary 18 any improvement past the Tamura value must use triple points — which is exactly the mechanism of Maiorana's step from 53 to 54.

[1] Indexed by the pair e of lines meeting there.

11 Equality-branch obstructions

When Δ is small the admissible degeneracy patterns form a short list, and saturation of Lemma 11 forces the multiplicities m_i of double-extreme vertices at consecutive direction classes to satisfy

$$m_{i-1} + m_i = 2s_i, \qquad 0 \le m_i \le s_i s_{i+1}, \qquad m_i \in \mathbb{Z}, \tag{4}$$

cyclically in the class sizes $(s_1, \ldots, s_R)$. For odd R the rational solution is unique; for even R solvability requires $\sum_i (-1)^i 2s_i = 0$.

Theorem 31 ($n = 14$)**.** *No arrangement of* 14 *lines with exactly three parallel pairs, or one parallel triad, and with no three lines concurrent admits an admissible family of* 54 *triangles.*

Proof. Paired layout: sizes are three 2's and eight 1's, $R = 11$ odd, and the unique solution of (4) violates an interval constraint for each of the $\binom{11}{3} = 165$ placements. Triad layout: one 3 and eleven 1's, $R = 12$ even, and the alternating sum is $\pm 4 \ne 0$ at each of the 12 placements. Both enumerations are certified by an exact integer program (pure standard library, no floating point). ☐

Theorem 32 ($n = 18$)**.** *No arrangement of* 18 *lines with exactly three parallel pairs, or one parallel triad, and with no three lines concurrent admits an admissible family of* 94 *triangles.*

Proof. Same mechanism with $R = 15$ and $R = 16$: all $\binom{15}{3} = 455$ paired placements violate an interval constraint (gap-parity classes $140 + 90 + 225$, 19 dihedral orbits with exact multiplicities, reflection quotients 231 and 9 all asserted by the certificate), and all 16 triad placements have alternating sum ± 4. ☐

Proposition 33 (the method stops at $n = 20$)**.** *At $n = 20$, $T = 117$ we have $\Delta = 9$ and $2Q = 6$, leaving three units of slack, so* (4) *must be relaxed by a defect vector, and feasible defects exist: for the paired layout ($R = 17$) a defect of* 2 *at one doubled class with run lengths $(u, 0, 14 - u)$, e.g.* $(6, 0, 8)$*; for the triad layout ($R = 18$) a defect of* 4 *with all $m_i = 1$. Hence no argument using only* (4) *can exclude $T = 117$ at $n = 20$.*

12 The two open small cases, and how to close them

Table 2 collects what is proved. Theorem 4 simplifies the picture decisively: there is only one quantity above a_3^s, and every target may be searched in the face convention without discarding parallels or multiple points.

Reading the table with simple-only bounds and the incomplete proof of (1) discarded leaves exactly two open cases below $n = 18$. At $n = 11$ the value is settled [11]; at $n = 13$ the peer-reviewed bound $\lfloor 13 \cdot 11/3 \rfloor = 47$ equals the construction. At $n = 12$,

$$38 \le K(12) \le 40,$$

and at $n = 14$,

$$54 \le K(14) \le 56,$$

where the lower endpoints are exact-rational certificates and the upper endpoints are the general-arrangement bound stated in [8]. The first monotone frontier at each n is therefore:

- $(12, 39)$: unsatisfiability proves $K(12) = 38$; satisfiability, after straight-line realisation and exact verification, gives a new lower bound 39 and leaves only $(12, 40)$.

- $(14, 55)$: unsatisfiability proves $K(14) = 54$; satisfiability gives a 55-triangle record and leaves only $(14, 56)$.

Neither unsatisfiability conclusion uses (1) or [8].

The current search uses the direct-gap formulation of Lemma 24: the audited crossing/parallel, concurrency and line-order axioms remain, while sidedness, polygon-overlap and crossing auxiliaries disappear. The exact face budget is retained. Per-line capacity counters give instances with 288,179 variables / 653,828 clauses at $(12, 39)$ and 858,782 / 1,891,433 at $(14, 55)$; omitting only those accelerator counters, while retaining the global face budget, gives sound solver-diversity instances of 45,014 / 164,537 and 128,612 / 424,240. The exact direct-gap predicate agrees with the independent strict-sign oracle on 109,104 nondegenerate triples in 4,000 arrangements across five degeneracy modes, with zero mismatches. SAT controls remain satisfiable at every known optimum $n = 5, \dots, 9$ and refute the next value; $(9, 22)$ fell from 388 seconds in the first face-only formulation to 111 seconds, and a direct-gap $(7, 12)$ DRAT is independently verified.

An orthogonal accelerator instantiates Lemma 25. At $(12, 39)$ the independently established values $K(11) = 32$ and $K(10) = 25$ give every one-line and two-line deletion cut; together with the vertex-sector clauses this produces 374,381 variables and 830,030 clauses. On the known $(9, 22)$ UNSAT control, the same two cut types reduced a same-process CaDiCaL run from 1,034,027 conflicts / 116.3 seconds to 164,802 / 19.5 seconds, while $(9, 21)$ remained satisfiable.

The shared-ray implication in Lemma 25 is available as a separate opt-in propagation cut. It adds 11,880 clauses and no variables at $(12, 39)$, giving 374,381 variables / 841,910 clauses. The exact-rational audit covered 83,115 face-sector assignments, 17,325 pairs of faces sharing two lines, and 1,591 same-ray cases in 3,600 arrangements; every same-ray case had the required remote concurrency. An exact six-line control realizes all four sectors at one line pair. Performance was mixed: a CaDiCaL $(8, 16)$ control improved from 164,869 to 153,108 conflicts, its $(9, 22)$ control was neutral, and Kissat's $(9, 22)$ control worsened. The cut therefore remains opt-in; its live $(12, 39)$ run is solver diversity, not a speedup or a verdict.

The endpoint form of the same argument is stronger. For two faces sharing only line r, a concurrency at one pair of side endpoints and equal ray directions forces the other endpoint concurrency; if both endpoints coincide, the apex-side bits are opposite. The exact replay now covers 73,349 single-line face pairs, 1,607 colliding endpoints, 84 colliding same-ray cases, and 42 shared segments, again with zero violations. Together with the shared-pair edge-parity specialization this adds 154,440 clauses and no variables at $(12, 39)$. The perturbation item of Lemma 25, instantiated with $\bar{a}_3^s(12) = 37$, reifies multipoint incidence and requires two incident selected faces; it adds 656 variables / 7,036 clauses. The direct $K(4) = 2$ clauses add another 1,980 clauses. With the existing 11- and 10-line deletion cuts, the resulting discovery instance has 375,037 variables / 1,005,366 clauses. Adding the checked $K(9) = 21$ deletion cuts gives 842,317 / 1,917,926.

Fresh same-process CaDiCaL controls show a genuine but nonuniform tradeoff. At $(8, 16)$ the shared-ray baseline used 249,513 conflicts / 25.2 seconds; endpoint closure used 221,905 / 18.2, and endpoint plus $K(4)$ used 17.6 seconds. At $(9, 22)$ endpoint closure was slower (1,020,652 conflicts / 169.6 seconds versus 958,507 / 138.4). Both known-optimum controls $(8, 15)$ and $(9, 21)$ remain satisfiable. Accordingly the two new $(12, 39)$ lanes are solver-diversity experiments, not a general speedup claim. They have no verdict and no proof logging.

Two riskier formulations were rejected as defaults. Exact reification of every face literal accepted the exact $n = 10$ and $n = 12$ witnesses, but improved $(8, 16)$ while making both $(9, 21)$ and $(9, 22)$ much slower. A zero-aware rank-three Grassmann–Plücker encoding passed 1,386,000 exact rational relations with no violation, but found no semantic gap in the base relaxation through $n = 7$ and doubled the $(8, 16)$ endpoint runtime. It is retained only as an opt-in propagation experiment.

n	constr.	$\bar{a}_3^s$	(1)	FK	status
6	7	7	7	8	$K(6) = 7$ certified here (DRAT)
8	15	14	15	16	$K(8) = 15$ certified here (DRAT)
9	21	21	21	21	$K(9) = 21$; FK attained
10	25	25	26	26	$K(10) = 25$ certified here (cube cover)
11	32	32	33	33	$K(11) = 32$ [11]
12	**38**	37	39	40	$[38, 40]$; $39 \Rightarrow$ multipoint (Thm. 34)
13	47	47	47	47	$K(13) = 47$; FK attained
14	**54** (verified)	53	55	56	$[54, 56]$; $(14, 55)$ running
18	93	93	95	96	$[93, 96]$
20	116	116	119	120	$[116, 120]$

Table 2: Constructions are lower bounds for $K = K_{\text{gen}}$ (Theorem 4). Column $\bar{a}_3^s$ is the exact maximum over *simple* arrangements [1, 2]: where the construction exceeds it (bold: $n = 8, 12, 14$) the record is attained only non-simply, so the simple-only bound $\lfloor n(n - 7/3)/3 \rfloor$ — which equals the construction at every even n here and is what the OEIS upper-bound column reports — cannot apply (Proposition 8). The reference columns are the claim (1) from the incomplete unpublished draft [3], and FK $= \lfloor n(n - 2)/3 \rfloor$ [8] (see Remark 2). Rows marked *certified here* are settled by unsatisfiability with every degeneracy free, citing nothing.

There is a smaller exact branch cut. With no finite multipoint, (2) gives $Q \leq 1$ at both frontiers. The $Q = 0$ branch is simple and already excluded by [2]. For $Q = 1$, rotation and relabelling put the unique parallel class at $\{0, 1\}$, so one canonical direct-gap cube covers the remaining no-concurrency branch. It has 20,895 variables / 111,753 clauses at $(12, 39)$ and 44,399 / 245,822 at $(14, 55)$.

Theorem 34 ($n = 12$ no-concurrency reduction)**.** *Every arrangement of* 12 *lines with* 39 *triangular faces has a finite point incident with at least three lines.*

Proof. Suppose there is no finite multipoint. Equation (2) gives $2Q \leq 120 - 3 \cdot 39 = 3$, hence $Q \leq 1$. If $Q = 0$, the arrangement is simple and [2, Thm. 3] gives at most 37 triangular faces. If $Q = 1$, rotate the circular slope order so the unique parallel class is $\{0, 1\}$. The primary direct-gap instance has 20,895 variables and 111,753 clauses, with no further symmetry assumption. Kissat reports UNSAT and `drat-trim` verifies its 2,437,053,136-byte proof: 19,802 formula clauses and 6,539,084 lemmas lie in the core, using 412,704,063 resolution steps. Independently, the valid global-sign symmetry $X(0, 2, 3)$ gives a 111,754-clause instance whose 759,328,450-byte proof is also verified (126,043,830 resolution steps). The hash-bound dual-certificate manifest and both checker transcripts are recorded in `scratch/kobon/n12/n12_q1_gap_certificate.json`. □

13 Open problems

1. **Decide the first frontiers** $(12, 39)$ **and** $(14, 55)$**.** By Theorem 4 each is one crossing-free instance with all degeneracies free. Unsatisfiability proves $K(12) = 38$ or $K(14) = 54$ respectively; satisfiability gives a new record and leaves the final rung $(12, 40)$ or $(14, 56)$.

2. **Find the right global degeneracy correction.** The coefficient-1 square penalty is false by

Proposition 15. More generally the family $\mathcal{R}_m$ rules out every universal inequality

$$3F \le n(n-2) - 2Q - c\sum_p (k_p - 2)^2$$

with fixed $c > 2/3$: the admissible ratio tends to $2/3$ from above. Proposition 26 certifies the endpoint at $n = 7$, but the general case remains open. Decide whether $c = 2/3$ is universal, or whether the correction must use nonlocal incidence data. Any valid positive correction would force simplicity at $\Delta = 0$. After Theorem 4, a T-member family can be replaced by T distinct triangular faces, giving $T \le F(\mathcal{A}) \le \bar{a}_3^s(n)$; equality between T and the total face count is not needed. Such an argument would provide a rigorous route to the prior mod-6 improvement, not a new numerical bound.

3. **Defect-tolerant obstructions.** Proposition 33 localises the missing ingredient: a constraint on where a defect may sit.

4. **Global structure of saturated flowers.** Proposition 12 shows that the $2a_\ell$ tokens of Lemma 11 are simultaneously realisable at each individual multipoint. Proposition 15 goes further: networks of saturated triple sectors can defeat the full square charge globally. Any replacement must measure how these networks close rather than sum a pointwise penalty.

5. **Pointwise structure.** Theorem 4 equates the two maxima but not the optima: crossed optima exist (§3). Which arrangements admit *only* crossed optima of maximum size, and can the descent be made canonical?

A Artefacts and replay

Release bundle `math/kobon/release/kobon-2026-08/` with `MANIFEST.sha256`; documentation hub `math/kobon/docs/`.

- `verification/maiorana_verify.py` — fail-closed certification of Theorem 28 (exit-0 transcript included).
- `verification/sweep_all_verify.py` — all fifteen witnesses.
- `verification/max_packing.py` — exact overlap graph and SAT optimality, used for Theorem 30.
- `scratch/kobon/broad_ladder.py` — ladder prober used for Theorem 27; instances `ladder_n8_t16.cnf`, `ladder_n9_t22.cnf` and the DRAT proofs.
- Direct-gap generator `gap_faces.py`, including hereditary, vertex-sector, endpoint-closure, perturbation, exact-face and projective chirotope options; exact 109,104-triple oracle comparison `gap_face_verify.py`; and the expanded exact endpoint replay `sector_bound_audit.py`.
- `scratch/kobon/reified_face_audit.py` — fixed exact-witness acceptance for the combined local-cut formulation.
- `chirotope_gp_audit.py` — determinant-sign and 1,386,000-relation Grassmann–Plücker replay.

- `scratch/kobon/frontier_endpoint_research.json` — hash-bound theorems, source survey, clause deltas, A/B controls, live-lane inputs and promotion gates for the endpoint phase.
- `scratch/kobon/pappus_relaxation_probe.py` — a checked non-Pappus model showing where field-specific incidence cuts can strengthen the abstract order relaxation; no Pappus clauses enter the current frontier.
- `scratch/kobon/q1_gap_faces.py` and `n12/n12_q1_gap_certificate.json` — generator, hash manifest and checker transcript for Theorem 34.
- `scratch/kobon/flower_counterexamples.py` — exact rational $k = 3, 4$ replays for Proposition 12.
- `scratch/kobon/square_penalty_counterexample.py` and its JSON output — exact rational $A(12, 1)$ replay for Proposition 15, with two face predicates and complete edge-use accounting.
- `scratch/kobon/square_penalty_sat.py` — weighted-penalty generator for Proposition 26.
- `c23_n7_violation.cnf`, its DRAT proof and checker log, and `c23_n7_certificate.json` — checked endpoint certificate.
- `proofs/appendix_independent_check.py`, `proofs/appendix_check_n18.py` — exact integer certificates for Theorems 31 and 32.
- `proofs/defect_lemma.md` — analysis behind Proposition 33.
- `scripts/engine.py` — exact rational geometry core and CNF builder.

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[17] Companion technical document `math/kobon/paper_capacity.md`: full case analysis of Lemma 11, audit protocols, refutations of weaker penalty forms, and the complete experiment ledger.